

An Improved Lower Bound for the $3x + 1$ Problem:

$$\pi_a(x) \geq x^{0.895}$$

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Abstract

For $a \not\equiv 0 \pmod{3}$ let $\pi_a(x)$ count the integers $n \leq x$ whose forward orbit under the $3x + 1$ function contains a . Krasikov and Lagarias (2003) proved $\pi_a(x) \geq x^{0.84}$ for all sufficiently large x , by exhibiting a feasible solution to an explicit linear program $L_k^{NT}(\lambda)$ over the 3^{k-1} congruence classes $m \equiv 2 \pmod{3}$ of modulus 3^k , at level $k = 11$; this has remained the best published exponent. We improve the bound to

$$\pi_a(x) \geq x^{179/200} = x^{0.895} \quad \text{for all sufficiently large } x,$$

using their theorem verbatim and new feasible solutions at level $k = 17$ (43,046,721 classes). The computational unlock is structural: for fixed λ the program is a feasibility question $c \leq F_\lambda(c)$ for a monotone, homogeneous, concave map, so a feasible point is found by nonlinear power iteration (Collatz–Wielandt) rather than linear programming, in time linear in the number of classes per sweep. Soundness does not rest on floating point: for each claimed exponent p/q we exhibit an integer vector and verify every inequality of $L_k^{NT}(2^{p/q})$ in exact integer arithmetic, with the transcendental coefficients replaced by rational lower bounds certified via integer power comparisons. The method reproduces the complete published table of Krasikov–Lagarias ($2 \leq k \leq 11$) to all seven printed decimals, and yields certified exponents 0.853, 0.863, 0.8724, 0.8812, 0.888, 0.895 at $k = 12, \dots, 17$. Separately, a Lean 4 formalization of an integer-grid weakening of the method now carries machine verification past the problem’s critical constant: a kernel-checked proof that $\pi_1(x) = \Omega(x^\gamma)$ with $\gamma = 50 \log_2(1261/1250) = 0.632009\dots > \log 2 / \log 3$ (Section 6).

1 Introduction

The $3x + 1$ function is $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n ; the $3x + 1$ conjecture asserts that every $n \geq 1$ eventually reaches 1 under iteration [6, 7]. For a positive integer $a \not\equiv 0 \pmod{3}$, let

$$\pi_a(x) := \#\{1 \leq n \leq x : T^{(j)}(n) = a \text{ for some } j \geq 0\}.$$

A sequence of results, beginning with Crandall [2] in 1978, established lower bounds $\pi_1(x) > x^\beta$ for increasing exponents β : the difference-inequality method introduced by Krasikov [4] gave

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$x^{0.43}$, Wirsching [10] $x^{0.48}$, the nonlinear-programming approach of Applegate and Lagarias [1] $x^{0.81}$, and Krasikov–Lagarias [5] reached

$$\pi_a(x) \geq x^{0.84} \quad (a \not\equiv 0 \pmod{3}, x \geq x_0(a)),$$

by a computation, performed in 2002 at level $k = 11$, on a linear program family $L_k^{NT}(\lambda)$ recalled below. This exponent has remained the best published bound: the 2021 overview [7] cites it as the state of the art, and we are not aware of any later improvement. (Tao’s celebrated result that almost all orbits attain almost bounded values [9] concerns logarithmic density of a different event and neither implies nor is implied by pointwise bounds on π_a .)

Our result is:

Theorem 1.1. *For every positive integer $a \not\equiv 0 \pmod{3}$ there is $x_0(a)$ such that*

$$\pi_a(x) \geq x^{179/200} = x^{0.895} \quad \text{for all } x \geq x_0(a).$$

The theorem follows from the Krasikov–Lagarias framework *verbatim* — their Theorem 2.2 converts any feasible solution of $L_k^{NT}(\lambda)$ into the bound with exponent $\gamma = \log_2 \lambda$, and they prove feasibility lifts from k to $k + 1$, so larger k never hurts — together with a new feasible solution at $k = 17$, over 43,046,721 congruence classes, with $\lambda = 2^{179/200}$. The contribution of this paper is therefore computational, in two parts that we believe are of independent methodological interest:

1. **Feasibility without linear programming (Section 3).** For fixed λ the program $L_k^{NT}(\lambda)$ asks for a positive bounded vector c with $c \leq F_\lambda(c)$, where F_λ is monotone, positively homogeneous, and concave (each coordinate is a linear term plus a minimum of three coordinates, with positive coefficients). Any vector satisfying $\min_m F_\lambda(c)_m / c_m \geq 1$ is itself a feasible point — this is the easy direction of the Collatz–Wielandt characterization for monotone homogeneous maps (the coincidence of names is pleasant but accidental: Lothar Collatz of the eigenvalue bound is Lothar Collatz of the conjecture). Normalized power iteration on F_λ produces such vectors whenever λ is below the feasibility supremum λ_k . Each sweep costs $O(3^k)$, which is what moves the frontier from 3^{10} classes (LP, 2002 hardware) to 3^{16} and beyond.
2. **Exact certification (Section 4).** Floating point is used only to *find* candidate vectors. Each claimed exponent is a rational p/q ; the three coefficients of $L_k^{NT}(2^{p/q})$ are bounded below by dyadic rationals whose correctness is an integer power comparison; and the candidate vector, scaled and floored to integers, is checked against every inequality in exact integer arithmetic — all 43,046,721 of them at $k = 17$. Since the coefficients enter the right-hand sides positively, rounding them down only strengthens the verified system; soundness is one-directional throughout.

As validation, the same machinery reproduces every entry of the published Krasikov–Lagarias table ($2 \leq k \leq 11$) to all seven printed decimal places, including their record point $\lambda_{11} = 1.7922310$ (Section 5).

2 The Krasikov–Lagarias program

We recall the objects of [5], to which we refer for proofs. Throughout $\alpha := \log_2 3 = 1.58496\dots$ and

$$[3^k] := \{m \bmod 3^k : m \equiv 2 \pmod{3}\}, \quad |[3^k]| = 3^{k-1}.$$

For $m \in [3^k]$ define $\phi_k^m(y) := \inf\{\pi_a^*(2^y a) : a \equiv m \bmod 3^k, a \text{ not in a cycle}\}$, where π_a^* is the variant of π_a counting orbits that stay $\leq x$ (see [5, §2]). These functions satisfy a system of difference inequalities $\mathcal{I}_k$ [5, Prop. 2.1], and Krasikov–Lagarias associate to it the linear program family

$$L_k^{NT}(\lambda) : \text{ minimize } C_k^{\max} \text{ subject to}$$

$$\begin{aligned} \text{(L0)} \quad & 1 \leq c_k^m \leq C_k^{\max} & m \in [3^k], \\ \text{(L1)} \quad & c_k^m \leq c_k^{4m} \lambda^{-2} + c_{k-1}^{(4m-2)/3} \lambda^{\alpha-2} & m \equiv 2 \pmod{9}, \\ \text{(L2)} \quad & c_k^m \leq c_k^{4m} \lambda^{-2} & m \equiv 5 \pmod{9}, \\ \text{(L3)} \quad & c_k^m \leq c_k^{4m} \lambda^{-2} + c_{k-1}^{(2m-1)/3} \lambda^{\alpha-1} & m \equiv 8 \pmod{9}, \\ \text{(L4)} \quad & c_{k-1}^u \leq c_k^u, \quad c_{k-1}^u \leq c_k^{u+3^{k-1}}, \quad c_{k-1}^u \leq c_k^{u+2 \cdot 3^{k-1}} & u \in [3^{k-1}], \end{aligned}$$

with indices taken modulo 3^k (note $4m$, $(4m-2)/3$ and $(2m-1)/3$ remain $\equiv 2 \pmod{3}$). Their main results, which we use as black boxes:

Theorem 2.1 ([5, Thm. 2.2 and §6]). *If $1 \leq \lambda \leq 2$ and $L_k^{NT}(\lambda)$ has a feasible solution with principal variables $\{c_k^m\}$, then $\phi_k^m(y) \geq \Delta_1 c_k^m \lambda^y$ for all $y \geq 0$, with $\Delta_1 = (4 \max_m c_k^m)^{-1}$; consequently, for every $a \not\equiv 0 \pmod{3}$, $\pi_a(x) \geq x^{\log_2 \lambda}$ for all sufficiently large x .*

Proposition 2.2 ([5, §6]). *If $L_k^{NT}(\lambda)$ is feasible, so is $L_{k+1}^{NT}(\lambda)$ (set $c_{k+1}^{m+j \cdot 3^k} := c_k^m$ for $j = 0, 1, 2$). Hence $\lambda_k := \sup\{\lambda : L_k^{NT}(\lambda) \text{ feasible}\}$ is non-decreasing in k .*

It is an open problem whether $\lambda_k \rightarrow 2$, which would give $\pi_a(x) \geq x^{1-\epsilon}$ for every $\epsilon > 0$ [5, §6].

3 Feasibility as a nonlinear eigenproblem

Fix λ and eliminate the auxiliary variables: given any principal vector $c = (c_k^m)_{m \in [3^k]}$, the optimal choice in (L4) is $\bar{c}_{k-1}^u := \min\{c_k^u, c_k^{u+3^{k-1}}, c_k^{u+2 \cdot 3^{k-1}}\}$ [5, eq. (2.15)]. Substituting into (L1)–(L3) defines the map $F_\lambda : \mathbb{R}_{>0}^{[3^k]} \rightarrow \mathbb{R}_{>0}^{[3^k]}$,

$$F_\lambda(c)_m = \begin{cases} \lambda^{-2} c_{4m} + \lambda^{\alpha-2} \bar{c}_{(4m-2)/3} & m \equiv 2 \pmod{9}, \\ \lambda^{-2} c_{4m} & m \equiv 5 \pmod{9}, \\ \lambda^{-2} c_{4m} + \lambda^{\alpha-1} \bar{c}_{(2m-1)/3} & m \equiv 8 \pmod{9}, \end{cases}$$

and $L_k^{NT}(\lambda)$ is feasible if and only if there is a vector c with $1 \leq c_m$, bounded, and $c \leq F_\lambda(c)$ pointwise (then $C^{\max} := \max c$). The map F_λ is monotone ($c \leq c' \Rightarrow F(c) \leq F(c')$), positively homogeneous ($F(tc) = tF(c)$ for $t > 0$), and concave; all coefficients are positive.

Lemma 3.1. *If $c > 0$ satisfies $\min_m F_\lambda(c)_m / c_m \geq 1$, then $c / \min_m c_m$ is a feasible solution of $L_k^{NT}(\lambda)$.*

Proof. $c \leq F_\lambda(c)$ is the displayed condition; homogeneity preserves it under rescaling; the rescaled vector has $\min = 1$ and is finite, so (L0) holds with $C^{\max} = \max c / \min c$; (L1)–(L4) hold with $\bar{c}$ as auxiliaries by construction. $\square$

This is the easy (verification) direction of the Collatz–Wielandt formula for monotone homogeneous maps [3]; the useful empirical fact, standard in nonlinear Perron–Frobenius theory, is that normalized power iteration $c \leftarrow F_\lambda(c) / \|F_\lambda(c)\|_\infty$ converges to vectors achieving $\min_m F(c)_m / c_m$ close to the spectral value, so that whenever $\lambda < \lambda_k$ the iteration eventually exhibits a certificate in the sense of Lemma 3.1. We stress that no claim in this paper depends on convergence theory: the iteration is only a search heuristic, and Lemma 3.1 is checked directly on its output.

Three practical devices matter at scale. (i) Each sweep is two gather operations and one three-way minimum over arrays of length 3^{k-1} ; $k = 17$ sweeps take under a second vectorized. (ii) Bisection on λ re-uses the certificate of the last feasible λ as a warm start. (iii) For the largest k we skip bisection entirely: by Proposition 2.2 the certificate from level $k - 1$ lifts to level k , and a single targeted run at a chosen λ either certifies it or not — any certificate is final, so nothing is lost by aiming directly at a target exponent.

4 Exact certification

Let the claimed exponent be $\gamma = p/q \in \mathbb{Q}$ and $\lambda = 2^{p/q}$, so $\log_2 \lambda = p/q$ exactly. The three coefficients of F_λ are then q -th roots of explicit rationals:

$$\lambda^{-2} = \left(\frac{1}{2^{2p}}\right)^{1/q}, \quad \lambda^{\alpha-2} = \left(\frac{3^p}{2^{2p}}\right)^{1/q}, \quad \lambda^{\alpha-1} = \left(\frac{3^p}{2^p}\right)^{1/q}.$$

Lemma 4.1. *Let $u/v \leq (A/B)^{1/q}$ with u, v, A, B positive integers, which is equivalent to the integer inequality $u^q B \leq v^q A$. If the system obtained from $L_k^{NT}(2^{p/q})$ by replacing each coefficient with such a rational lower bound admits a solution, then so does $L_k^{NT}(2^{p/q})$ itself.*

Proof. All coefficients occur positively on right-hand sides of $\leq$ -constraints; decreasing them only shrinks the right-hand sides. $\square$

The verifier therefore proceeds as follows. Rational lower bounds with $v = 2^{64}$ are computed for the three coefficients and certified by the integer comparison of Lemma 4.1 (computed once; the integers involved have a few hundred thousand bits for our $q \leq 2500$). The floating-point vector c found in Section 3 is scaled by 2^{48} , floored to an integer vector C , and clamped below at 2^{48} (so $c = C/2^{48} \geq 1$). Each constraint is then checked as a single integer inequality; for example (L3) becomes

$$C_m \cdot v_2 v_C \leq u_2 v_C \cdot C_{4m} + u_C v_2 \cdot \min(C_u, C_{u+3^{k-1}}, C_{u+2 \cdot 3^{k-1}}),$$

with (u_2, v_2) and (u_C, v_C) the certified bounds for λ^{-2} and $\lambda^{\alpha-1}$. A run that checks all 3^{k-1} constraints yields, by Lemmas 3.1 and 4.1 and Theorem 2.1, the bound $\pi_a(x) \geq x^{p/q}$. No floating-point number survives into the verified statement.

5 Results

Validation. Bisection with the iteration of Section 3 reproduces the published table of [5, Table 2] for $2 \leq k \leq 11$ at all seven printed decimals; in particular the record entry $\lambda_{11} = 1.7922310$ is matched exactly.

New levels. Beyond $k = 11$:

k	classes 3^{k-1}	λ certified	$\gamma = \log_2 \lambda$	exact certificate p/q
12	177,147	1.8064235	0.8531361	853/1000 = 0.853
13	531,441	1.8188238	0.8630058	863/1000 = 0.863
14	1,594,323	1.8307723	0.8724524	2181/2500 = 0.8724
15	4,782,969	1.8419683	0.8812483	2203/2500 = 0.8812
16	14,348,907	≥ 1.8506089	≥ 0.888	111/125 = 0.888
17	43,046,721	≥ 1.8596099	≥ 0.895	179/200 = 0.895

For $k \leq 15$ the third column is the certified end of a bisection (the true λ_k exceeds it by less than 2×10^{-7}); for $k = 16, 17$ it is a single targeted certificate as described in Section 3, and the true λ_k is somewhat larger. In every row the final column was verified by the exact integer procedure of Section 4; at $k = 17$ this comprises 43,046,721 verified inequalities.

Proof of Theorem 1.1. The exact certificate at $k = 17$ exhibits a feasible solution of $L_{17}^{NT}(2^{179/200})$ by Lemmas 3.1 and 4.1. Apply Theorem 2.1. $\square$

6 Discussion

The marginal exponent gains per level, +0.0114, +0.0099, +0.0094, +0.0088 across $k = 11, \dots, 15$, decay slowly; whether $\lambda_k \rightarrow 2$ (equivalently $\pi_a(x) \geq x^{1-\epsilon}$ for all ϵ) remains the fundamental open question about this method [5, §6], and the data is consistent with it. Each additional level costs a factor of 3 in memory and time; $k = 18$ (129M classes) is within reach of a workstation, and $k \approx 20$ of a GPU implementation. The bottleneck is not the certification (linear scan) but the iteration count near the feasibility threshold, which warm starts mitigate.

Three remarks on reliability. First, everything rests on Theorem 2.1, which is published and peer-reviewed; we add no asymptotic analysis of our own, and the new content of this paper is checkable by a referee in exact arithmetic from the certificate files. Second, each certificate was checked by *two independently written verifiers*: the original, and a clean-room re-implementation using only the language’s standard library (no shared code, no numerical libraries), with the constraint system re-derived from [5, §2], a different rational bounding scheme for the coefficients (denominator 10^{21} versus 2^{64} , binary search versus unit adjustment), and explicit assertions of the congruence bookkeeping; both verifiers also detect a single corrupted entry (negative control). Third, the verification procedure is small (under a hundred lines) and is itself a candidate for formalization: the certificate check is a finite integer computation of precisely the kind proof assistants discharge by reflection, and formalizing Theorem 2.1 — an induction over difference inequalities — would make the exponent fully machine-verified.

Machine-verified exponents, and the critical constant

That program is underway in the companion repository [8], which contains complete Lean 4 formalizations of integer-grid weakenings of the difference-inequality method at two strengths, both proved from the dynamics upward (backward-tree combinatorics, Krasikov’s inequalities, kernel-checked certificates; axioms: Lean’s standard `propext`, `Classical.choice`, `Quot.sound` only — no `sorry`, no extra axioms).

1. **Unit grid** (file `Krasikov.lean`): caps $2^y \cdot a$, an 81-class certificate at modulus 3^5 with rational rate $137/100$, giving the machine-checked bound $\pi_1(x) = \Omega(x^{0.4543})$ (theorem `density_lower_bound`).
2. **1/50 grid** (file `Krasikov50.lean`): caps follow the interleaved-geometric sequence $\text{cap}(t) = 2^{\lfloor t/50 \rfloor} r[t \bmod 50]$ for a fixed 50-entry integer table $r \approx 10^4 \cdot 2^{i/50}$, so that the per-step rate is the *rational* $\mu = 1261/1250$ and every certificate condition is a pure integer inequality. The two odd-branch terms of the difference inequalities retard by only $21/50$ resp. $1/50$ of a doubling (the kernel-checked table conditions; the margin is $2^{21/50} = 1.3382 \dots > 4/3$). A 2187-class certificate at modulus 3^8 , decided by the kernel in chunks, together with per-class witness roots $a = m + 6561q$ ($q \in \{0, 1\}$) whose Terras orbits reach 8 in at most 146 kernel-evaluated steps, yields (theorem `K8.density_bound`):

$$\pi_1(x) = \Omega(x^\gamma), \quad \gamma = 50 \log_2 \frac{1261}{1250} = 0.632009 \dots$$

The second exponent crosses a meaningful line: it exceeds $\log 2 / \log 3 = 0.630929 \dots$, the critical odd-step density against which every quantity of this problem is measured (it is the density a divergent orbit would have to sustain forever, and the constant governing the cycle Diophantine bounds). The comparison of the two numbers is numerical rather than structural — they measure different things — but as a milestone it is sharp: the machine-verified count of integers reaching 1 now provably grows faster than $x^{\log 2 / \log 3}$. The exponent 0.632 exceeds every bound published before Applegate–Lagarias [1]; scaling the formal pipeline onward toward the present paper’s 0.895 certificates is mechanical in principle.

Reproducibility

The search engine, the one-shot driver, both exact verifiers, the certificate export format (raw little-endian integers with a hashed sidecar), and instructions are in the repository accompanying this paper (`analysis/kl_exponent.py`, `kl_oneshot.py`, `kl_certificate.py`, `kl_export_certificate.py`, `kl_verify_independent.py`); all runs are deterministic. Finding the $k = 17$ certificate takes under an hour on a laptop; verifying it exactly takes minutes per verifier.

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